



Release 4

[. \(http://hl7.org/fhir/R4/searchform.html\)](http://hl7.org/fhir/R4/searchform.html)

[HL7 \(http://www.hl7.org\)](http://www.hl7.org)

[Conformance \(conformance-module.html\)](#) > **CompartmentDefinition**

This page is part of the FHIR Specification (v4.0.1: R4 - Mixed **Normative** (<https://confluence.hl7.org/display/HL7/HL7+Balloting>) and **STU** (<https://confluence.hl7.org/display/HL7/HL7+Balloting>)) in it's permanent home (it will always be available at this URL). The current version which supercedes this version is **5.0.0** (<http://hl7.org/fhir/index.html>). For a full list of available versions, see the [Directory of published versions](#) (<http://hl7.org/fhir/directory.html>). Page versions: **R5** (<http://hl7.org/fhir/R5/compartmentdefinition.html>) **R4B** (<http://hl7.org/fhir/R4B/compartmentdefinition.html>) **R4** **R3** (<http://hl7.org/fhir/STU3/compartmentdefinition.html>)

Content	Examples (compartmentdefinition-examples.html)	Detailed Descriptions (compartmentdefinition-definitions.html)	Mappings (compartmentdefinition-mappings.html)
Profiles & Extensions (compartmentdefinition-profiles.html)	R3 Conversions (compartmentdefinition-version-maps.html)		

5.6_Resource CompartmentDefinition - Content

FHIR Infrastructure  (http://www.hl7.org/Special/committees/fiwg/index.cfm), Work Group	Maturity Level (versions.html#maturity): 1	Trial Use (versions.html#std-process).	Security Category (security.html#SecPrivConsiderations): Anonymous	Compartments (compartmentdefinition.html) : Not linked to any defined compartments
---	--	--	---	--

A compartment definition that defines how resources are accessed on a server.

5.6.1 Scope and Usage

Each resource may belong to one or more logical compartments. A compartment is a logical grouping of resources which share a common property. Compartments have two principal roles:

- Function as an access mechanism for finding a set of related resources quickly
- Provide a definitional basis for applying access control to resources quickly

Note:

At present, compartment definitions can *only* be defined by HL7 International. This is because their existence creates significant impact on the behavior of servers.

5.6.2 Boundaries and Relationships

Compartment definitions describe how particular compartment instances are named and identified, and how systems know which resources are in the compartment. The following compartments are defined by this specification:

Title	Description	Identity	Membership
Patient (compartmentdefinition-patient.html)	The set of resources associated with a particular patient	There is an instance of the patient compartment for each patient resource, and the identity of the compartment is the same as the patient. When a patient is linked to another patient, all the records associated with the linked patient are in the compartment associated with the target of the link.	The patient compartment includes any resources where the subject of the resource is the patient, and some other resources that are directly linked to resources in the patient compartment
Encounter (compartmentdefinition-encounter.html)	The set of resources associated with a particular encounter	There is an instance of the encounter compartment for each encounter resource, and the identity of the compartment is the same as the encounter	The encounter compartment includes any resources where the resource has an explicitly nominated encounter, and some other resources that themselves link to resources in the encounter compartment. Note that for many resources, the exact nature of the link to encounter can be ambiguous (e.g. for a DiagnosticReport, is it the encounter when it was initiated, or when it was reported?)
RelatedPerson (compartmentdefinition-relatedperson.html)	The set of resources associated with a particular 'related person'	There is an instance of the relatedPerson compartment for each relatedPerson resource, and the identity of the compartment is the same as the relatedPerson	The relatedPerson compartment includes any resources where the resource is explicitly linked to relatedPerson (usually as author)
Practitioner (compartmentdefinition-practitioner.html)	The set of resources associated with a particular practitioner	There is an instance of the practitioner compartment for each Practitioner resource, and the identity of the compartment is the same as the Practitioner	The practitioner compartment includes any resources where the resource is explicitly linked to a Practitioner (usually as author, but other kinds of linkage exist)
Device (compartmentdefinition-device.html)	The set of resources associated with a particular device	There is an instance of the device compartment for each Device resource, and the identity of the compartment is the same as the Device	The device compartment includes any resources where the resource is explicitly linked to a Device (mostly subject or performer)

The full definitions of these compartments are published as **CompartmentDefinition** resources. Servers typically do not support the full definition of a compartment, and are not required to. Systems may publish **CompartmentDefinition** resources so that other systems may make use of compartments properly.

- CompartmentDefinitions are used by **CapabilityStatement** ([capabilitystatement.html](#)) instances for specifying how resources are accessed

This resource is referenced by **CapabilityStatement** ([capabilitystatement.html#CapabilityStatement](#)).

5.6.3 Resource Content

Structure

UML

XML

JSON

Turtle

R3 Diff

All

Structure

Name (formats.html#table)	Flags (formats.html#table)	Card. (formats.html#table)	Type (formats.html#table)	Description & Constraints (formats.html#table)
 CompartmentDefinition (compartmentdefinition-definitions.html#CompartmentDefinition)	I (conformance-rules.html#constraints) TU (versions.html#std-process)		DomainResource (domainresource.html)	 Warning: Name should be used as identifier for the module by many processing applications such as generation Elements defined in Ancestors: (resource.html#Resource) , met (resource.html#Resource) , imp (resource.html#Resource) , lang (resource.html#Resource) , text (domainresource.html#DomainResource) , contained (domainresource.html#DomainResource) , modifierExtension (domainresource.html#DomainResource)
 url (compartmentdefinition-definitions.html#CompartmentDefinition.url)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	1..1	uri (datatypes.html#uri)	Canonical identifier for this compartment definition, represented as a URI (unique)
 version (compartmentdefinition-definitions.html#CompartmentDefinition.version)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	0..1	string (datatypes.html#string)	Business version of the compartment definition
 name (compartmentdefinition-definitions.html#CompartmentDefinition.name)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary) I (conformance-rules.html#constraints)	1..1	string (datatypes.html#string)	Name for this compartment definition (computer friendly)
 status (compartmentdefinition-definitions.html#CompartmentDefinition.status)	?I (conformance-rules.html#isModifier) Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	1..1	code (datatypes.html#code)	draft active retired unknown PublicationStatus (valueset-publication-status.html) (Required (terminologies.html#required))
 experimental (compartmentdefinition-definitions.html#CompartmentDefinition.experimental)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	0..1	boolean (datatypes.html#boolean)	For testing purposes, not real use
 date (compartmentdefinition-definitions.html#CompartmentDefinition.date)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	0..1	dateTime (datatypes.html#dateTime)	Date last changed
 publisher (compartmentdefinition-definitions.html#CompartmentDefinition.publisher)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	0..1	string (datatypes.html#string)	Name of the publisher (organization individual)
 contact (compartmentdefinition-definitions.html#CompartmentDefinition.contact)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	0..*	ContactDetail (metadataatypes.html#ContactDetail)	Contact details for the publisher
 description (compartmentdefinition-definitions.html#CompartmentDefinition.description)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	0..1	markdown (datatypes.html#markdown)	Natural language description of compartment definition
 useContext (compartmentdefinition-definitions.html#CompartmentDefinition.useContext)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	0..*	UsageContext (metadataatypes.html#UsageContext)	The context that the content is in support
 purpose (compartmentdefinition-definitions.html#CompartmentDefinition.purpose)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	0..1	markdown (datatypes.html#markdown)	Why this compartment definition is defined
 code (compartmentdefinition-definitions.html#CompartmentDefinition.code)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	1..1	code (datatypes.html#code)	Patient Encounter RelatedPerson Practitioner Device CompartmentType (valueset-compartment-type.html) (Required (terminologies.html#required))
 search (compartmentdefinition-definitions.html#CompartmentDefinition.search)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	1..1	boolean (datatypes.html#boolean)	Whether the search syntax is supported
 resource (compartmentdefinition-definitions.html#CompartmentDefinition.resource)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	0..*	BackboneElement (backboneelement.html)	How a resource is related to the compartment
 code (compartmentdefinition-definitions.html#CompartmentDefinition.resource.code)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	1..1	code (datatypes.html#code)	Name of resource type ResourceType (valueset-resource-types.html) (Required (terminologies.html#required))
 param (compartmentdefinition-definitions.html#CompartmentDefinition.resource.param)	Σ (elementdefinition-definitions.html#ElementDefinition.isSummary)	0..*	string (datatypes.html#string)	Search Parameter Name, or choice of parameters
 documentation (compartmentdefinition-definitions.html#CompartmentDefinition.resource.documentation)		0..1	string (datatypes.html#string)	Additional documentation about resource and compartment
 Documentation for this format (formats.html#table)				

See the [Profiles & Extensions \(compartmentdefinition-profiles.html\)](#) and the alternate definitions: Master Definition [XML \(compartmentdefinition.profile.xml.html\)](#) + [JSON \(compartmentdefinition.profile.json.html\)](#), [XML \(xml.html\)](#) Schema ([compartmentdefinition.xsd](#)) Schematron ([compartmentdefinition.sch](#)) + [JSON \(json.html\)](#) Schema ([compartmentdefinition.schema.json.html](#)), [ShEx \(compartmentdefinition.shex.html\)](#) (for [Turtle \(rdf.html\)](#)) + see the extensions ([compartmentdefinition-profiles.html](#)) & the [dependency analysis \(compartmentdefinition-dependencies.html\)](#)

5.6.3.1 Terminology Bindings

Path	Definition	Type	Reference
CompartmentDefinition.status	The lifecycle status of an artifact.	Required (terminologies.html#required)	PublicationStatus (valueset-publication-status.html)
CompartmentDefinition.code	Which type a compartment definition describes.	Required (terminologies.html#required)	CompartmentType (valueset-compartment-type.html)
CompartmentDefinition.resource.code	One of the resource types defined as part of this version of FHIR.	Required (terminologies.html#required)	Resource Types (valueset-resource-types.html)

5.6.3.2 Constraints

id	Level	Location	Description	Expression (fhirpath.html)
----	-------	----------	-------------	----------------------------

cpd-0	Warning (conformance-rules.html#warning)	(base)	Name should be usable as an identifier for the module by machine processing applications such as code generation	<code>name.matches(' [A-Z] ([A-Za-z0-9_]) {0,254}')</code>
-------	--	--------	--	--

5.6.3.3 Using Compartments

As an example of compartment usage, to retrieve a list of a patient's conditions, use the URL:

```
GET [base]/Patient/[id]/Condition
```

Additional search parameters can be defined, such as this hypothetical search for acute conditions:

```
GET [base]/Patient/[id]/Condition?code:in=http://hspc.org/ValueSet/acute-concerns
```

Note that as searches, these are syntactic variations on these two search URLs respectively:

```
GET [base]/Condition?patient=[id]
GET [base]/Condition?patient=[id]&code:in=http://hspc.org/ValueSet/acute-concerns
```

The outcome of a compartment search is the same as the equivalent normal search. For example, both these searches return the same outcome if there is no patient 333:

```
GET [base]/Patient/333/Condition
GET [base]/Condition?patient=333
```

Whether the patient doesn't exist, or the user has no access to the patient, both these searches return an empty bundle with no matches. Some systems will include an operation outcome warning that there is no matching patient.

However, there is a key difference in functionality between compartment based searches and direct searches with parameters. Consider this search:

```
GET [base]/Patient/[id]/Communication
```

Because the definition of the [patient compartment \(compartmentdefinition-patient.html\)](#) for [Communication \(communication.html\)](#) says that a Communication resource is in the patient compartment if the subject, sender, or recipient is the patient, the compartment search is actually the same as the union of these 3 searches:

```
GET [base]/Communication?subject=[id]
GET [base]/Communication?sender=[id]
GET [base]/Communication?recipient=[id]
```

There is no way to do this as a single search, except by using the [filter \(search_filter.html\)](#):

```
GET [base]/Communication?_filter=subject re [id] or sender re [id] or recipient re [id]
```

Further details of searching by compartment are [described under the search operation \(http.html#vsearch\)](#). As a search related operation, the assignment of resources to compartments is only based on the current version of any of the resources involved. Note that [contained \(references.html#contained\)](#) patient resources cannot create a patient compartment of their own.

Note that while this specification describes how to use the compartment syntax to find resources that are logically associated with the compartment, the compartment is not part of the identity of the resource. E.g. the response to the following is not defined by this specification:

```
GET [base]/Patient/[patient-id]/Condition/[resource-id]
```

The response for write operations (PUT/DELETE/PATCH) are also not defined by this specification. Nor is the response to a POST defined:

```
POST [base]/Patient/[patient-id]/Condition
```

There is no expectation for servers to support either read or write to such URLs.

5.6.3.4 Logical Meaning of Compartments

Compartments may be used explicitly, like this, but can also be used implicitly. For instance, if a FHIR server is providing a patient view of a record, the authorized user associated with use of the FHIR RESTful API may be limited to accessing records from the compartment instance(s) logically associated with their identity. Irrespective of whether compartments are being used explicitly or implicitly, servers will need to make arrangements to make some resources with no direct link to a patient available to the client (medications, substances, etc.).

Note that resources may cross between compartments, or interlink them. Examples of this would be where a [Diagnostic Report \(diagnosticreport.html\)](#) identifies a subject, but an [Observation \(observation.html\)](#) it references identifies a different subject, or where a [List \(list.html\)](#) resource references items that identify different subjects. Such cross-linking may arise for many valid reasons, including:

- Cases where subject records are inter-linked - Transplants, Perinatal care, family therapy etc.
- Workflow management where action lists link multiple patients and/or practitioners

Given the wide variety of use cases and contexts in which FHIR is used, compartments do not define how cross-linking is handled. Systems may reject resources, remove them from both compartments, or place them in both, or act in some other fashion.

The [graph definition \(graphdefinition.html\)](#) resource provides a method by which rules about cross-linking may be made and enforced.

It is at the discretion of the server whether to include resources in a compartment when the reference to the resource that establishes the compartment is in an extension.

Some resources are not in any compartment, e.g. [Medication \(medication.html\)](#), [Substance \(substance.html\)](#), [Location \(location.html\)](#). These resources are not linked directly to a patient or authored record, and are sometimes called 'master files'. Servers will need to make arrangements to make these resources available to the clients that are limited to particular compartments. For example, a Medication resource describes a medication itself and does not link to a patient; however, a resource such as MedicationAdministration connects the Medication (details of what was administered) to the patient (for whom was it administered), and so is required to interpret the administration.

5.6.4_Defining New Compartments

Compartments are defined and added to the list above when implementer communities identify them as common access points for data. As described below, compartments have both syntactical and logical consequences, and both these aspects of their functionality are evaluated when deciding whether to define compartments.

5.6.5 Search Parameters

Search parameters for this resource. The [common parameters \(search.html#all\)](#) also apply. See [Searching \(search.html\)](#) for more information about searching in REST, messaging, and services.

Name	Type	Description	Expression	In Common
code	token (search.html#token)	Patient Encounter RelatedPerson Practitioner Device	CompartmentDefinition.code	
context	token (search.html#token)	A use context assigned to the compartment definition	(CompartmentDefinition.useContext.value as CodeableConcept)	
context-quantity	quantity (search.html#quantity)	A quantity- or range-valued use context assigned to the compartment definition	(CompartmentDefinition.useContext.value as Quantity) (CompartmentDefinition.useContext.value as Range)	
context-type	token (search.html#token)	A type of use context assigned to the compartment definition	CompartmentDefinition.useContext.code	
context-type-quantity	composite (search.html#composite)	A use context type and quantity- or range-based value assigned to the compartment definition	On CompartmentDefinition.useContext: context-type: code context-quantity: value.as(Quantity) value.as(Range)	
context-type-value	composite (search.html#composite)	A use context type and value assigned to the compartment definition	On CompartmentDefinition.useContext: context-type: code context: value.as(CodeableConcept)	
date	date (search.html#date)	The compartment definition publication date	CompartmentDefinition.date	
description	string (search.html#string)	The description of the compartment definition	CompartmentDefinition.description	
name	string (search.html#string)	Computationally friendly name of the compartment definition	CompartmentDefinition.name	
publisher	string (search.html#string)	Name of the publisher of the compartment definition	CompartmentDefinition.publisher	
resource	token (search.html#token)	Name of resource type	CompartmentDefinition.resource.code	
status	token (search.html#token)	The current status of the compartment definition	CompartmentDefinition.status	
url	uri (search.html#uri)	The uri that identifies the compartment definition	CompartmentDefinition.url	
version	token (search.html#token)	The business version of the compartment definition	CompartmentDefinition.version	